Prevention of Nosocomial Infection — Full Notes

Extended study notes — chain of infection, standard precautions, and sterilization methods, expanded from the quick-reference version.

The chain of infection

- Infectious agent — the pathogen itself (bacteria, virus, fungus, prion)
- Reservoir — where the pathogen lives and multiplies (patients, equipment, water systems)
- Portal of exit — how it leaves the reservoir (respiratory droplets, blood, wounds)
- Mode of transmission — contact, droplet, airborne, vehicle-borne, vector-borne
- Portal of entry — mucous membranes, broken skin, invasive devices
- Susceptible host — anyone with reduced immunity, open wounds, or invasive lines
- Breaking any single link in this chain prevents transmission — hand hygiene alone breaks several links at once

Standard precautions (applied to every patient, every time)

- Hand hygiene before and after every patient contact
- Personal protective equipment matched to the expected exposure
- Safe handling and disposal of sharps
- Respiratory hygiene / cough etiquette
- Safe injection practices — one needle, one syringe, one time
- Environmental cleaning between patients and procedures

Transmission-based precautions (added on top of standard precautions)

- Contact precautions — gown and gloves, dedicated equipment where possible
- Droplet precautions — surgical mask within short range of the patient
- Airborne precautions — N95/FFP respirator, negative-pressure room where available

Sterilization vs. disinfection vs. cleaning

- Cleaning — physical removal of organic matter and debris; always the first step
- Disinfection — reduces microorganisms to a safe level, does not necessarily kill spores
- Sterilization — destroys all forms of microbial life including spores

Common sterilization methods

- Autoclaving (steam under pressure) — the most common method for reusable instruments
- Dry heat — for items that steam would damage but that tolerate high temperature
- Ethylene oxide gas — for heat-sensitive items, requires aeration time afterward
- Chemical/cold sterilization — for delicate items that can't tolerate heat or gas

Surgical site infection risk factors

- Patient factors — diabetes, obesity, smoking, immunosuppression, malnutrition
- Procedure factors — duration of surgery, degree of contamination, technique
- Environmental factors — theatre traffic, ventilation, instrument sterility